

Reinforcement Learning via Evolutionary Fine-Tuning Loops for LLM Debiasing

**Kandavel Yogeshram
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**Department of Electrical
Engineering and Computer Science**

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Abstract

The problem with Large Language Models (LLMs) is that they often mirror human bias present in their training dataset. Traditional alignment strategies like Reinforcement Learning from Human Feedback (RLHF) (Ouyang et al., 2022) and Direct Preference Optimization (DPO) (Rafailov et al., 2023) suffer from dataset limitations. In this paper, we propose **Reinforcement learning via Evolutionary Fine-Tuning Loops (REFTL)**, a fully automated, self-directed framework that is designed to reduce LLM bias while preserving general academic capabilities. **REFTL** uses a closed-loop system comprised of: (1) self-directed question generation with MinHash Locality-Sensitive Hashing (LSH) to prevent redundancy, (2) prompt-driven persona answering and self-evaluation filtering with automated penalty functions, (3) evolutionary optimization across multiple nodes, (4) multi-adaptor merging using Drop and Rescale - Trim Elect Sign and Merging (DARE-TIES) (Yadav et al., 2023; Yu et al., 2023). Evaluated on Meta Llama-3.2-1B-Instruct, REFTL achieves significant debiasing on the Bias Benchmark for QA (BBQ) (Parrish et al., 2022). With a scaling merge weight of 2.0, REFTL reduces the ambiguous context bias score from 0.04247 to 0.01764 while more than doubling average ambiguous context accuracy from 10.85% to 24.11% and increasing total accuracy from 29.01% to 32.97%. This comes with a modest alignment tax in disambiguated context accuracy (falling from 47.17% to 41.85%). Benchmark evaluation on Massive Multitask Language Understanding (MMLU) (Hendrycks et al., 2020) shows negligible accuracy change (36.55% vs. 36.61% baseline), confirming that REFTL effectively mitigates bias without degrading general academic capabilities.

1. Introduction

Large language models learn social bias present in society and as they are used in different contexts, these biases can lead to harm. (Parrish et al., 2022) Thus, the alignment of Large Language Models with human centered values like fairness, neutrality and non-bias is extremely important for using LLMs in different contexts.

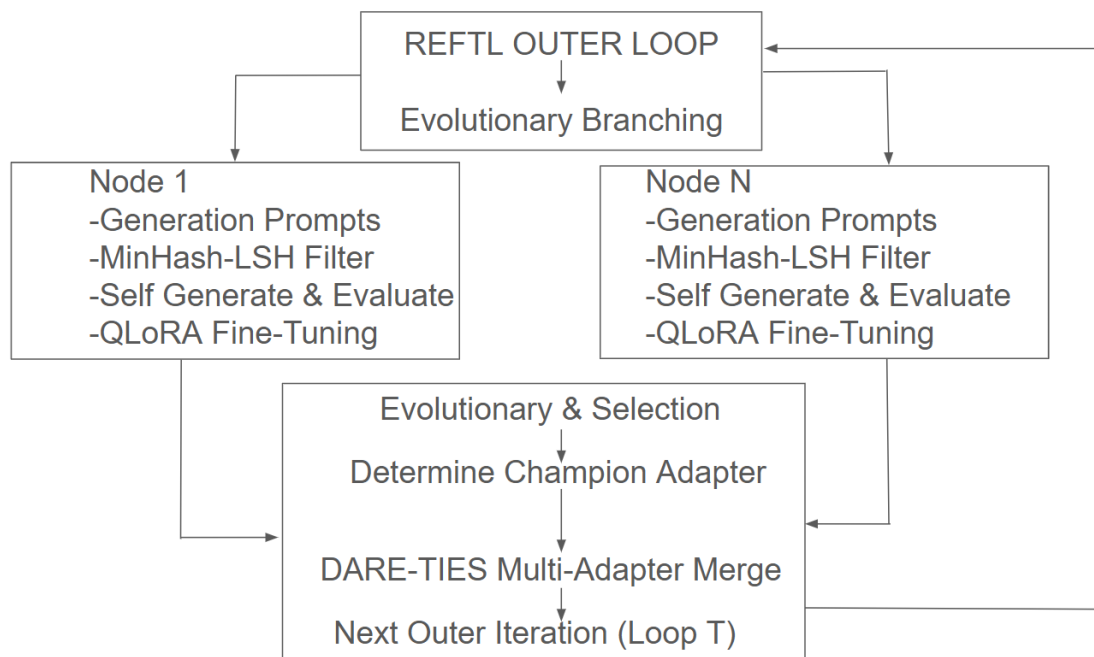
To align LLMs, standard methodologies rely on heavily curated datasets or relies on human feedback such as RLHF (Ouyang et al., 2022) and DPO (Rafailov et al., 2023)

However, these methods are limited by:

- **Limited Amount of Dataset:** There is a limit to how much dataset humans can produce.
- **Expense on Dataset Creation:** Creating large preference datasets is expensive when relying on human annotators, driving interest in Reinforcement Learning from AI Feedback (RLAIF) (Lee et al., 2023) to generate scalable datasets using LLMs.
- **Memory heavy:** Performing RLHF training requires a lot of space of memory on the GPU

To overcome these constraints, we introduce **Reinforcement Learning via Evolutionary Fine-Tuning Loops (REFTL)**. **REFTL** leverages Parameter-Efficient Fine Tuning (PEFT) and evolutionary algorithms to construct an automated. resource-efficient alignment pipeline. Rather than training a static model, REFTL treats the alignment task as performing evolutionary search of merged QLoRa (Dettmers et al., 2023) that is aligned.

Figure 1: REFTL Flow chart



Our system design splits the optimization into outer loops of 10 iterations and inner loop of 3 parallel evolutionary nodes. In each iteration, the model generates its own training questions, answers under debiasing persona instruction, filters the results through self-evaluation, fine-tunes QLoRa adapters (Dettmers et al., 2023), selects the best performing adapter path based on self-evaluation and penalty function, and merges the cumulative history of champion adapters using DARE-TIES. (Yadav et al., 2023; Yu et al., 2023).

Codes available at

<https://github.com/kandavely/Reinforcement-Learning-via-Evolutionary-Fine-Tuning-Loops-REFTL-Llama-3.2-1B-Instruct/tree/main>

2. Related Work

2.1. LLM Debiasing and Self-Alignment

Early debiasing research focused on prefix-tuning, token filtering, or post-hoc output modification. More recent frameworks employ constitutional design and self alignment.

Self-Instruct (Wang et al., 2022) demonstrates that models can generate their own instruction-tuning data. Constitutional AI (Bai et al., 2022) and automated feedback frameworks like UltraFeedback (Cui et al., 2023) leverage self-critique and model-generated preference feedback to align models with minimal human intervention. Similarly, Self-Rewarding Language Models (Yuan et al., 2024) show that models can act as their own reward judge to iteratively refine instruction-following performance. Recent frameworks like Self-Play Fine-Tuning (SPIN) (Chen et al., 2024) similarly demonstrate iterative self-alignment without external human supervision. REFTL builds upon this line of inquiry but introduces a closed-loop evolutionary selection process to guarantee that self-generated data actively optimizes downstream performance while keeping the data distribution diverse.

2.2 Parameter-Efficient Fine-Tuning (PEFT)

To avoid the costs of full-parameter tuning, LoRA (Hu et al., 2021) and QLoRA (Dettmers et al., 2023) freeze the base network parameters and injects trainable decomposition matrices.

The forward pass is modified as:

$$h = W_0x + BAx \quad (1)$$

where B and A are low rank adapters.

REFTL utilizes QLoRA (Dettmers et al., 2023) to fine-tune to dramatically reduce the GPU memory usage.

2.3 Model Merging and Interference Elimination

Combining specialized models in the parameter space has emerged as a low-cost alternative to multi-task learning, with recent work showing that evolutionary algorithms can automate parameter search across disparate base models (Akiba et al., 2024). While early weight-space ensembling techniques such as Model Soups (Wortsman et al., 2022) established that linear weight averaging across fine-tuned checkpoints can boost generalization, traditional weight averaging (WA) often causes parameter degradation due to sign conflicts and task interference when combining diverse tasks. Building upon the principles of task arithmetic (Ilharco et al., 2022), TIES-Merging (Yadav et al., 2023) resolves sign conflicts using parameter consensus. DARE (Yu et al., 2023) improves this by dropping a high proportion of fine grained parameters and rescaling the remaining weights to preserve model capabilities, making it ideal for combining multiple low-rank adapters over consecutive iterations.

3. Methodology

REFTL is structured around a continuous loop managing data generation, quality filtering, parameter-efficient optimization, and multi-adapter evolutionary selection.

3.1. Self-Directed Diversity-Enforced Data Generation

To prevent semantic collapse—where the LLM repetitively generates identical or highly similar training queries—we implement a Locality-Sensitive Hashing (LSH) engine based on MinHash (Broder, 1997), following modern LLM dataset deduplication practices (Lee et al., 2021).

A question is rejected and filtered if it matches any existing question within the LSH index above a threshold.

3.2. Self-Evaluation and Data Curation

For each accepted question, the model generates a response under a debiased persona instruction ("Fair and Unbiased Chatbot"). This yields the question and answer pair.

The model is then prompted under an expert evaluator persona to score the response quality in $[0, 100]$. To prevent structural degeneration (such as the model generating infinite repetitive lists or numerical templates to artificially inflate its score), we define a **Number Density Penalty Function**. The number density is found by dividing the total number of numbers with the length of the text.

3.3. Fine-Tuning and Parameter-Efficient Optimization

Using the curated dataset, we perform QLoRA fine-tuning on the base model. The model parameters are quantized into 4-bit NormalFloat (**nf4**) format. Fine-tuning updates only the injected low-rank matrices across all linear layers (**q_proj**, **k_proj**, **v_proj**, **o_proj**, **gate_proj**, **up_proj**, **down_proj**).

We employ the Autoregressive Sequence Fine-Tuning (SFT) loss over the formatted chat messages.

3.4. Evolutionary Selection and DARE-TIES Multi-Adapter Merging

To identify the optimal optimization path, the inner loop processes **N=3** parallel evolutionary nodes per iteration. Each node generates its own training trajectory and saves its resulting adapter weights.

At the end of an inner loop, each adapter merges with previous best adapters and is evaluated to obtain an average self-evaluation score. The node yielding the highest score is crowned the "champion" adapter for that generation **step t**.

To accumulate alignment knowledge across all successful generational checkpoints without increasing parameter count or causing interference, we merge the history of all champion adapters up to **step t** using PEFT's multi-adapter loading engine via the **DARE-TIES** consensus framework.

The DARE-TIES (Yadav et al., 2023; Yu et al., 2023) merge executes three operations:

1. **Drop (Pruning)**: Elements of each delta adapter are set to zero with probability $p = 1 - \text{density}$ (where density = 0.8 is the density parameter):

2. **Sign Consensus (TIES):** We determine the dominant parameter direction using sign voting.
3. **Consensus Masking and Rescaling:** We eliminate parameter values opposing the dominant sign direction, average the remaining parameters, and rescale the weights by $\frac{1}{density}$ to preserve task-arithmetic magnitude:

4. Experimental setup

4.1 Computational Infrastructure

All experiments were conducted on a dedicated local desktop computer. The detailed hardware configuration utilized for model training and evaluation is summarized in Table 1.

Table 1: Hardware specifications for the experimental environment.

Component	Specification
CPU	Intel(R) Core(TM) i7-10700 CPU @ 2.90GHz
Memory	16.0 GB RAM
GPU Acceleration	NVIDIA GeForce RTX 2060 SUPER

4.2 Model Configuration and Compute Runtime

We selected Meta Llama-3.2-1B Instruct (Grattafiori et al. ,2024) as the foundational model across all experimental iterations.

The end-to-end framework execution—encompassing automated dataset synthesis, evolutionary QLoRA (Detrmers et al., 2023) fine-tuning across parallel nodes, adapter merging and self evaluation—required an aggregate runtime of approximately 18 to 19 hours. To maintain long-term autonomous pipeline execution without human intervention or thermal throttling, an automated compute management routine was incorporated into the training pipeline. This routine actively monitored GPU thermal limits and introduced regulated pause intervals when temperatures surpassed safe operating thresholds, thereby preserving hardware integrity and operational consistency during extended optimization cycles.

5. Results and Evaluation

To comprehensively evaluate the effectiveness of the Reinforcement Learning via Evolutionary Fine-Tuning Loops (REFTL) framework, we benchmarked the baseline Meta Llama-3.2-1B-Instruct model (Grattafiori et al., 2024) against its REFTL-aligned variants across two primary evaluation dimensions:

1. **Bias Mitigation & Fairness:** Evaluated on the Bias Benchmark for QA (BBQ) (Parrish et al., 2022) across ambiguous and disambiguated context settings.
2. **General Capabilities Preservation:** Evaluated on Massive Multitask Language Understanding (MMLU) (Hendrycks et al., 2020) to determine whether alignment induces catastrophic forgetting or degradation in core reasoning.

We evaluated the base model against two merged adapter configurations: an evolutionary-merged LoRA model loaded with **a scaling weight of 1.0** and **a scaling weight of 2.0**.

5.1 BBQ Test Result

The BBQ (Parrish et al., 2022) benchmark evaluates model choices under ambiguous contexts (where relying on societal stereotypes indicates social bias) and disambiguated contexts (where factual evidence is explicitly provided). The parameter **target_loc** identifies the index of the multiple-choice option corresponding to the stereotyped group target. Ground-truth values for **target_loc** were extracted directly from the official BBQ dataset metadata repository (https://github.com/nyu-mll/BBQ/blob/main/supplemental/additional_metadata.csv; Parrish et al., 2022).

Summary of BBQ Performance

The empirical results across all model configurations on the BBQ benchmark are detailed in **Table 2**.

Table 2: Comparative BBQ Benchmark Metrics across Model Configurations.

Model Configuration	Total Accuracy (%)	Ambiguous Context Acc. (%)	Disambiguated Context Acc. (%)	Ambiguous Bias Score	Disambiguated Bias Score	Total Unanswered (%)
Base Model (Llama-3.2-1B)	29.01	10.85	47.17	0.04247	0.03425	0.01
REFTL (Merge Weight 1.0)	30.26	15.32	45.22	0.03379	0.03110	0.02
REFTL (Merge Weight 2.0)	32.97	24.11	41.85	0.01764	0.01882	0.10

Baseline Performance

The unmodified base model exhibits significant social bias and low accuracy in ambiguous context settings:

- **Accuracy:** Achieves an overall total accuracy of **29.01%**, with a low ambiguous context accuracy of **10.85%** and a disambiguated context accuracy of **47.17%**.

- **Bias Scores:** Yields an ambiguous context bias score of **0.04247** and a disambiguated context bias score of **0.03425**.
- **Unanswered & Target Location:** Shows minimal unanswerable responses (**0.01%**).

5.1.1 Impact of REFTL Alignment (Merge Weights 1.0 and 2.0)

Applying the REFTL framework yields consistent improvements in both bias reduction and overall accuracy, with scaling the merge weight to **2.0** producing the strongest debiasing effect:

- **Bias Score Reduction:**
 - At weight **1.0**, the ambiguous bias score drops from **0.04247** to **0.03379**, and the disambiguated bias score decreases to **0.03110**.
 - At weight **2.0**, the framework achieves a dramatic reduction in social bias: the ambiguous bias score declines to **0.01764**, while the disambiguated bias score falls to **0.01882**.
- **Accuracy Improvements & Alignment Tax:**
 - **Ambiguous Accuracy:** Total accuracy increases to **30.26%** (Weight 1.0) and **32.97%** (Weight 2.0). This overall gain is driven by a substantial surge in ambiguous context accuracy, rising from **10.85%** (Base) to **15.32%** (Weight 1.0) and **24.11%** (Weight 2.0)—more than doubling the base performance.
 - **Alignment Tax:** As anticipated in self-directed alignment frameworks, there is a minor trade-off in disambiguated context accuracy. Disambiguated accuracy experiences a modest decrease from **47.17%** (Base) to **45.22%** (Weight 1.0) and **41.85%**

(Weight 2.0). This slight decrease represents the expected 'alignment tax'—the performance trade-off on general capabilities often observed during safety and debiasing alignment (Askell et al., 2021).

- **Response Completeness:** Unanswered question percentages remain negligible across all configurations (**0.01%** to **0.10%**), demonstrating that REFTL debiasing does not cause refusal loops or output degradation.

5.1.2 BBQ Test result for Base model

Figure 2: BBQ Test accuracy result for Base model

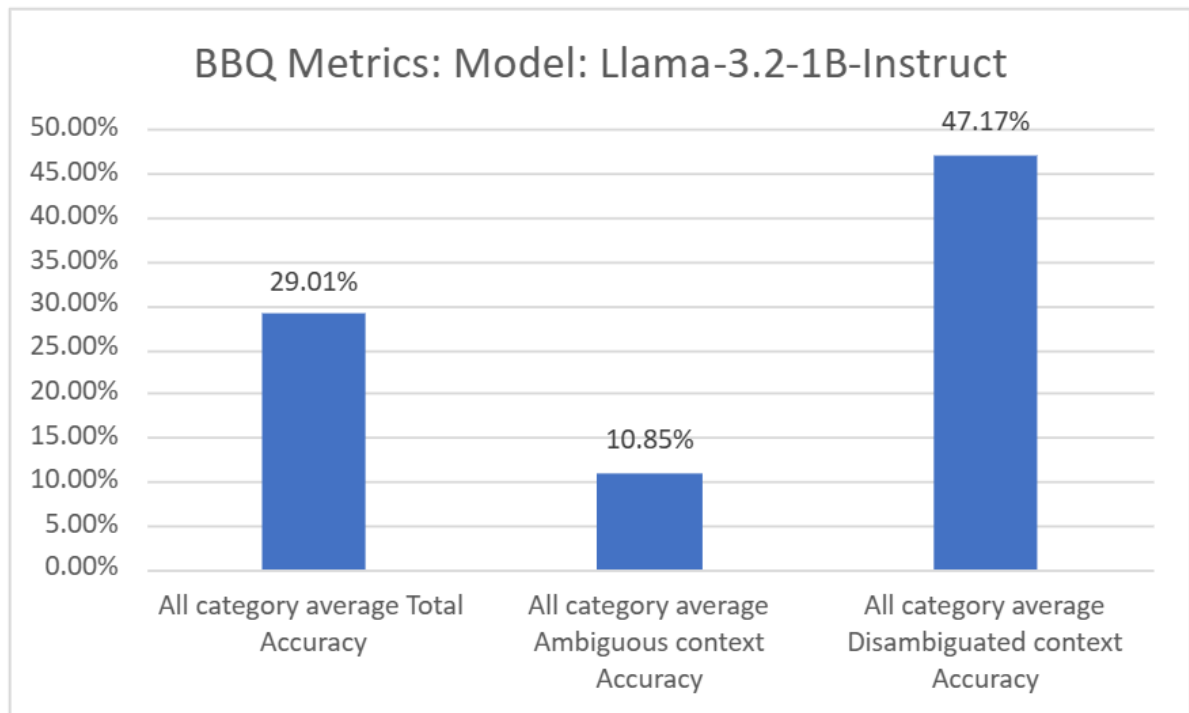


Figure 3: BBQ Test bias score result for Base model

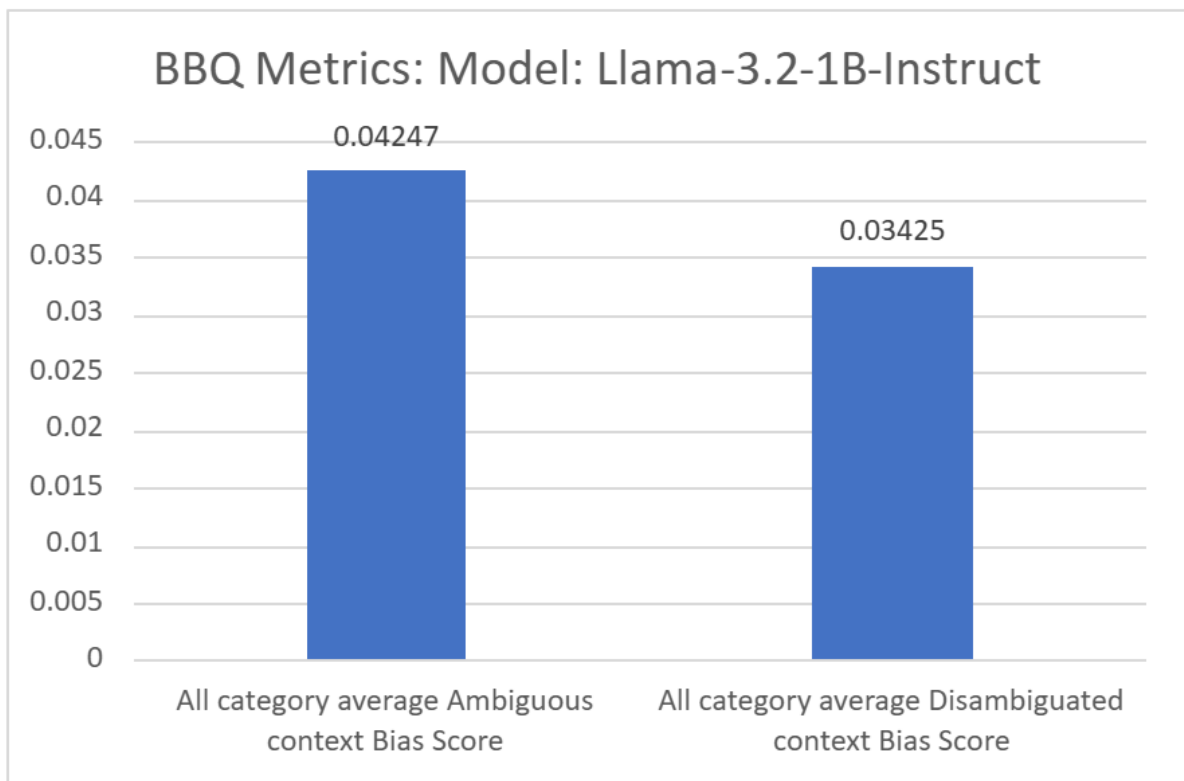


Figure 4: BBQ Test unanswered percentage for Base model

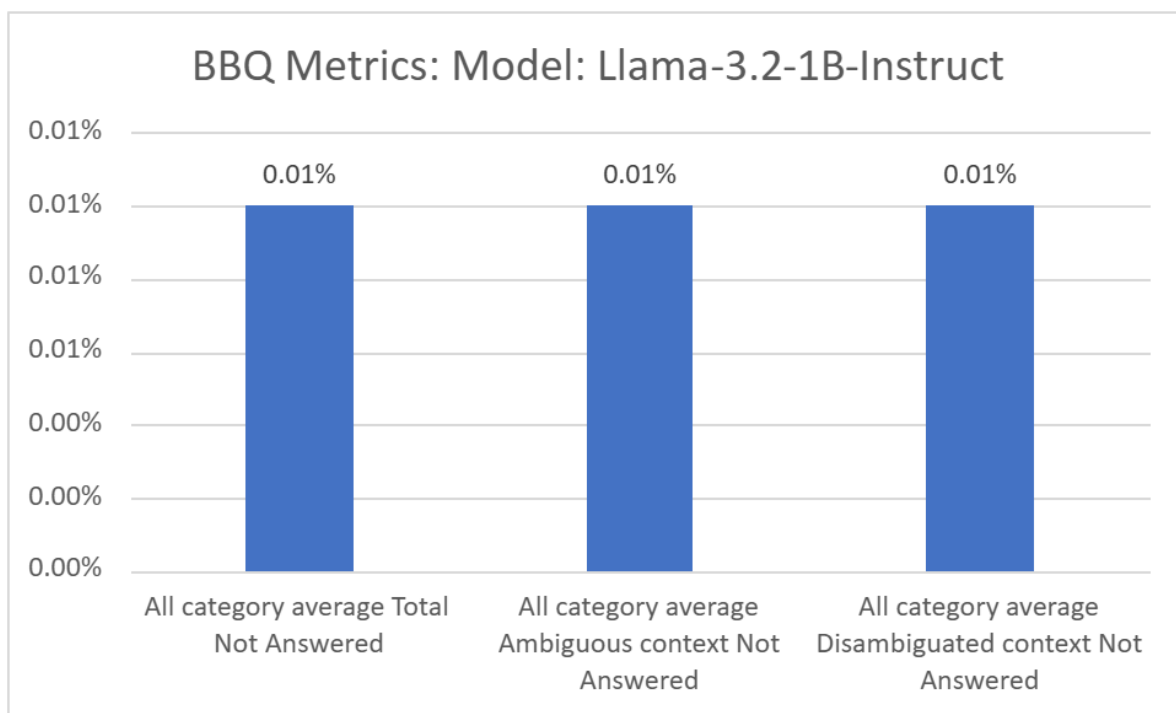
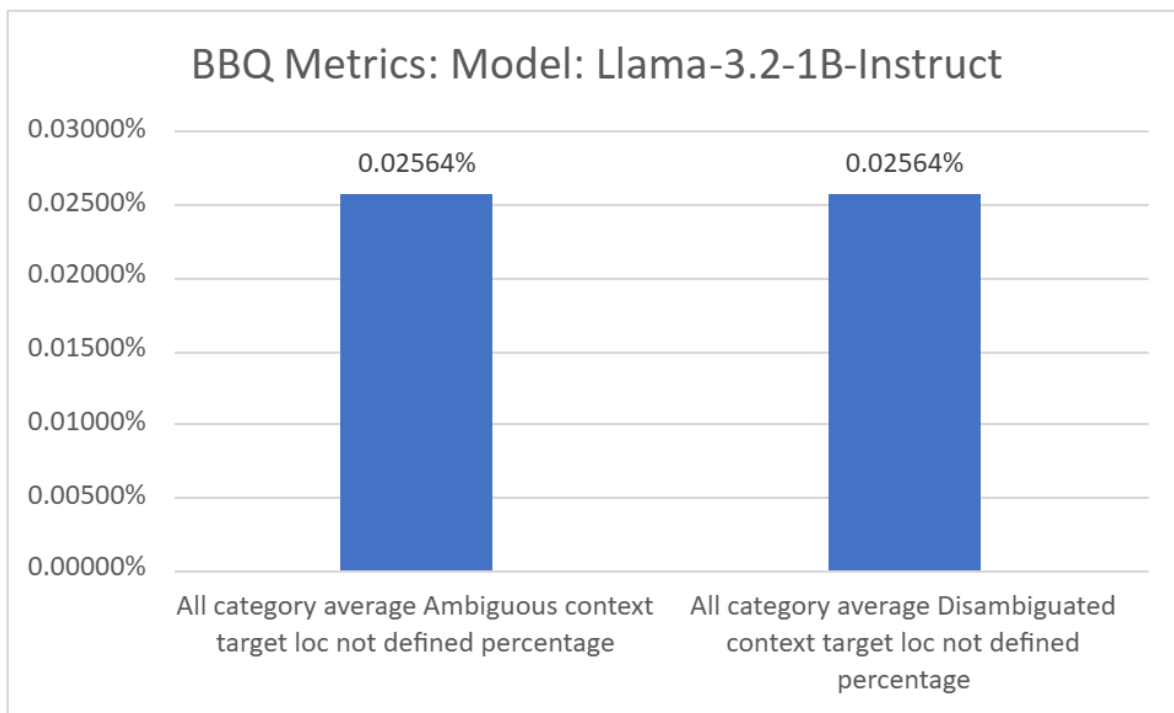


Figure 5: BBQ Test target loc not defined percentage for Base model



5.1.3 BBQ Test result for model loaded with merged LoRa adapters with 1.0 weight

Figure 6: BBQ Test accuracy result for model loaded with finetuned LoRas merged with weight 1.0

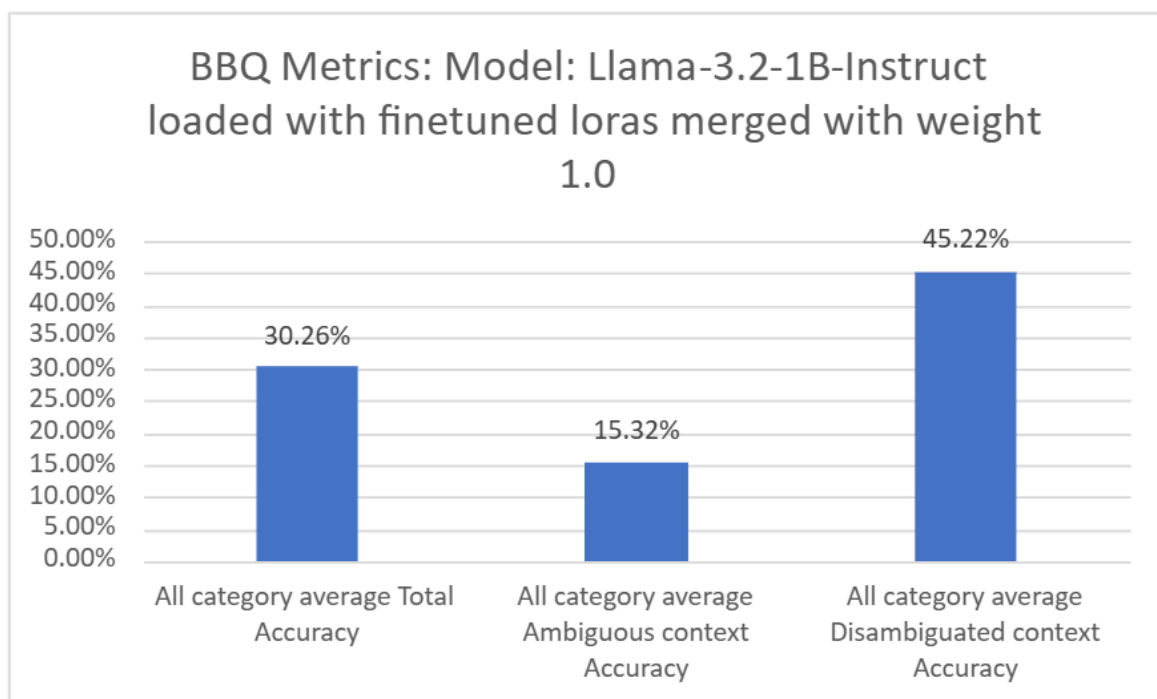


Figure 7: BBQ Test bias score result for model loaded with finetuned LoRas merged with weight 1.0

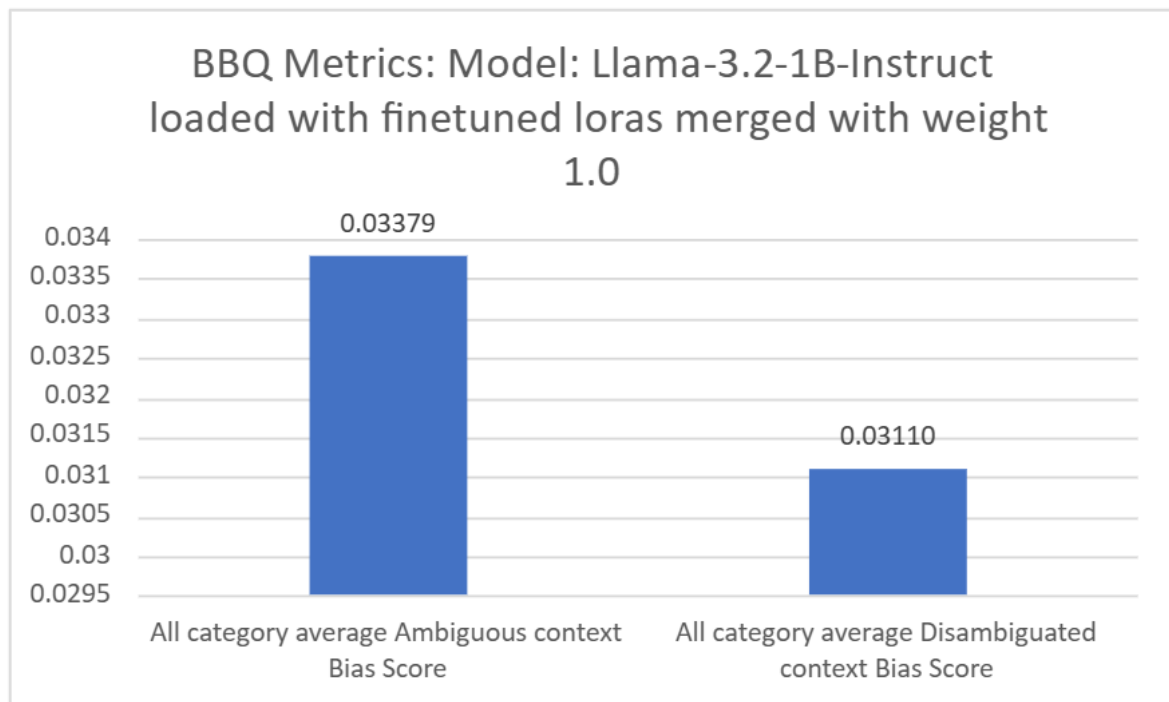


Figure 8: BBQ Test unanswered percentage for model loaded with finetuned LoRas merged with weight 1.0

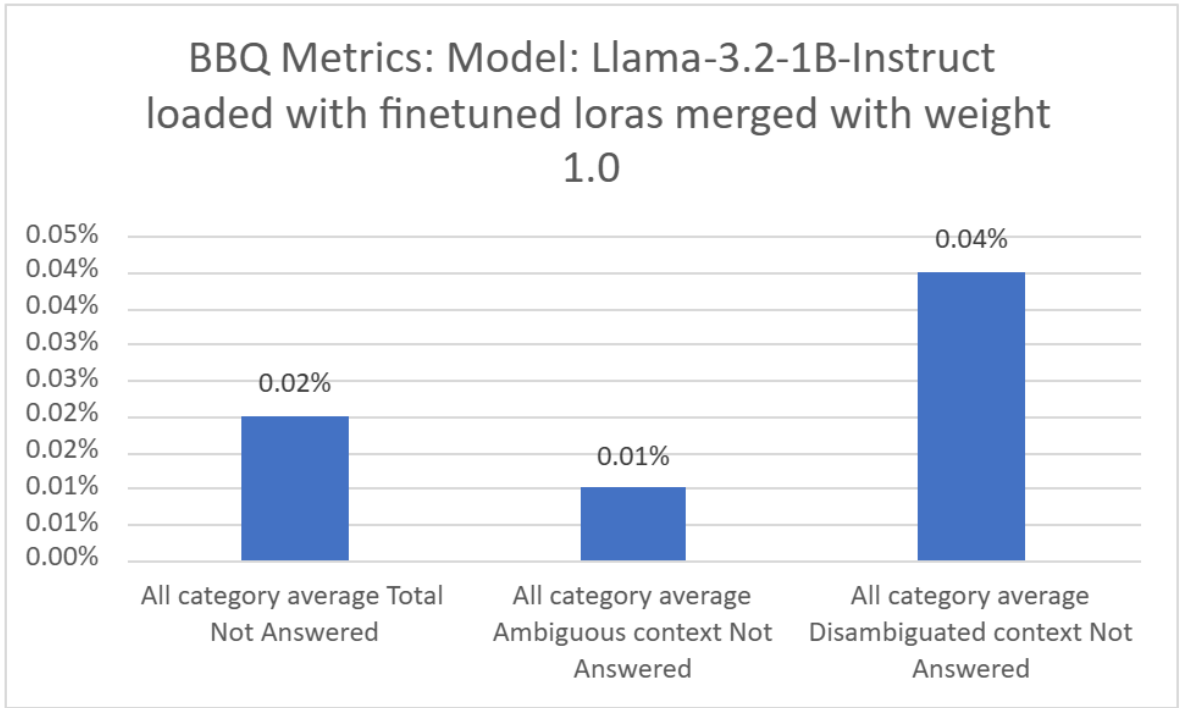
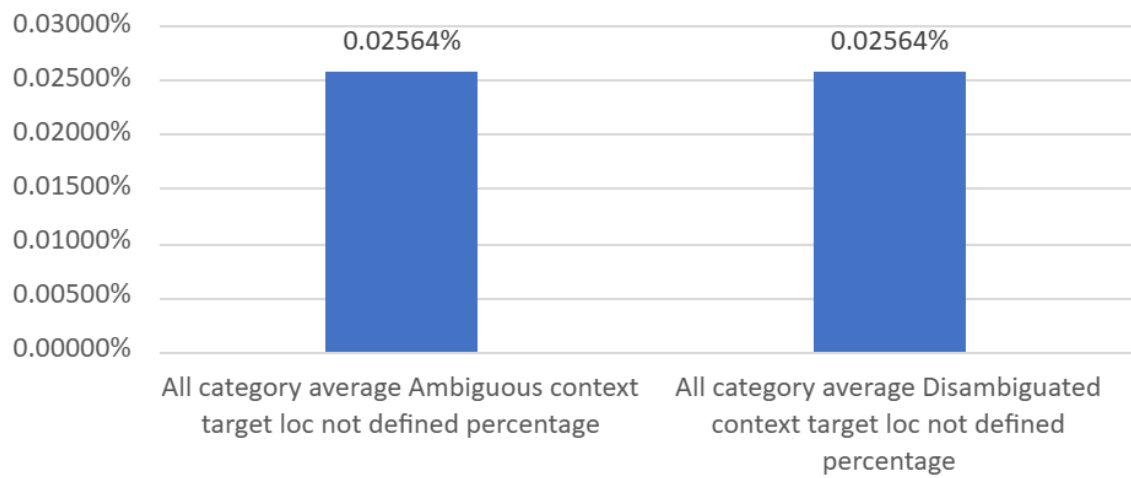


Figure 9: BBQ Test target loc not defined percentage for model loaded with finetuned LoRas merged with weight 1.0

BBQ Metrics: Model: Llama-3.2-1B-Instruct
loaded with finetuned loras merged with weight
1.0



5.1.4 BBQ Test result for model loaded with merged LoRa adapters with 2.0 weight

Figure 10: BBQ Test accuracy for model loaded with finetuned LoRas merged with weight 2.0

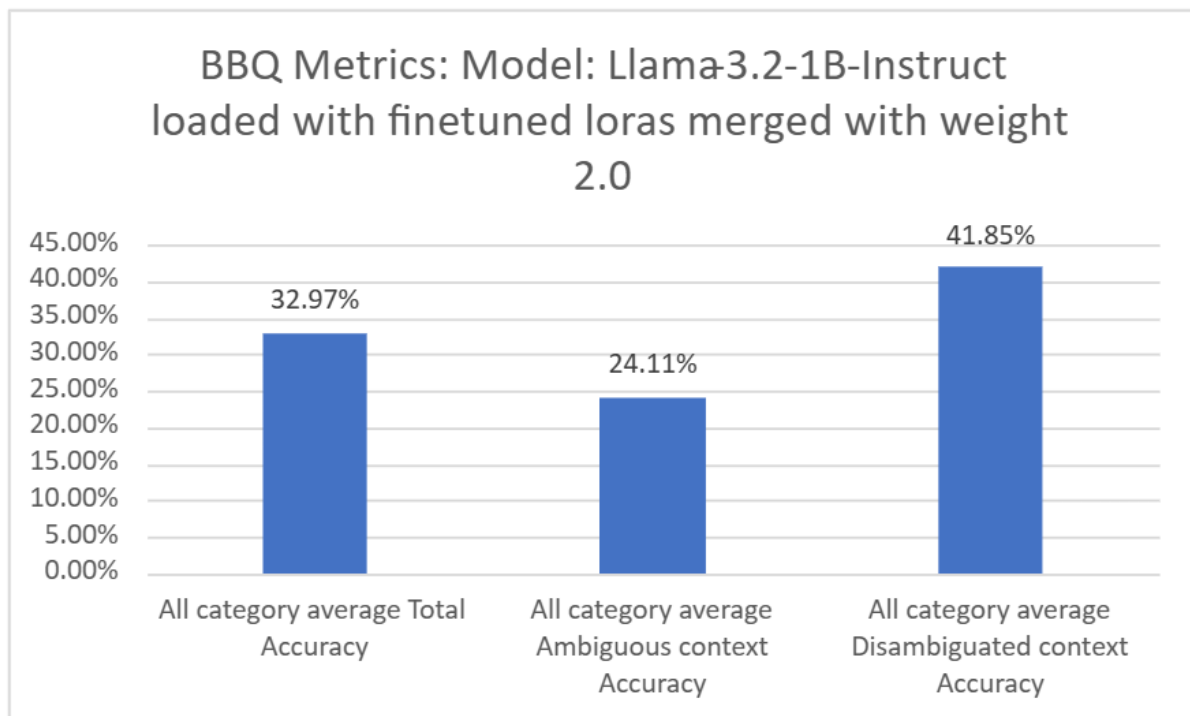


Figure 11: BBQ bias score for model loaded with finetuned LoRas merged with weight 2.0

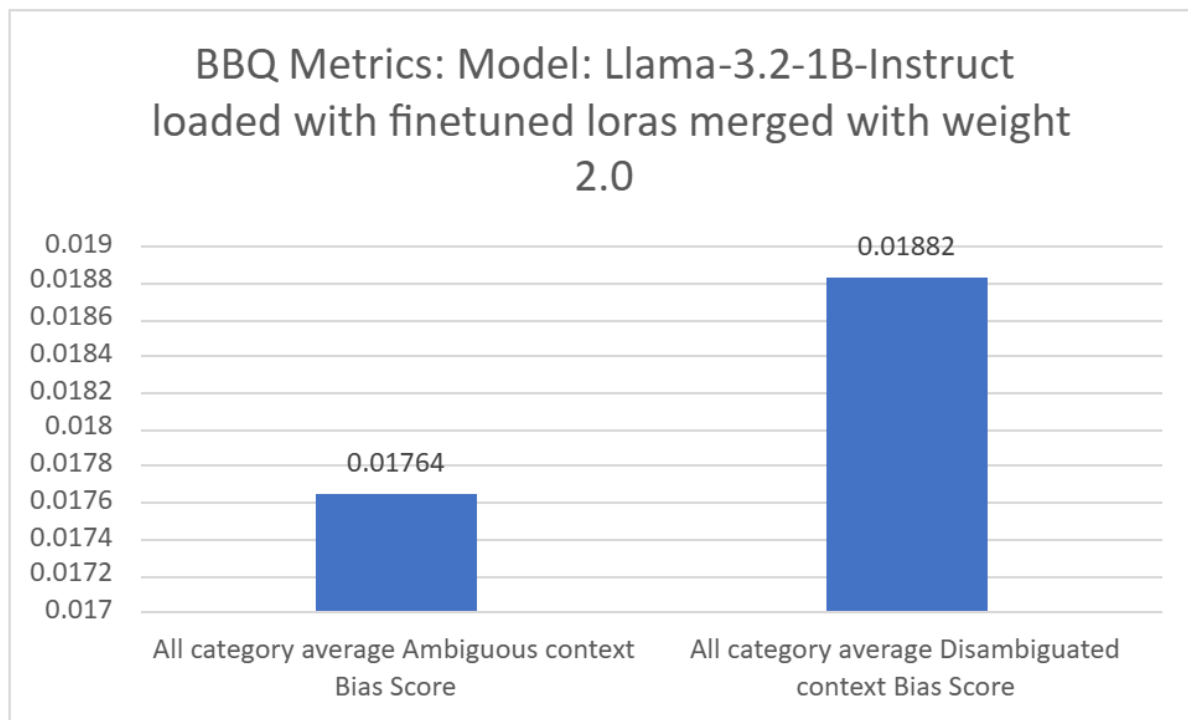


Figure 12: BBQ unanswered percentage for model loaded with finetuned LoRas merged with weight 2.0

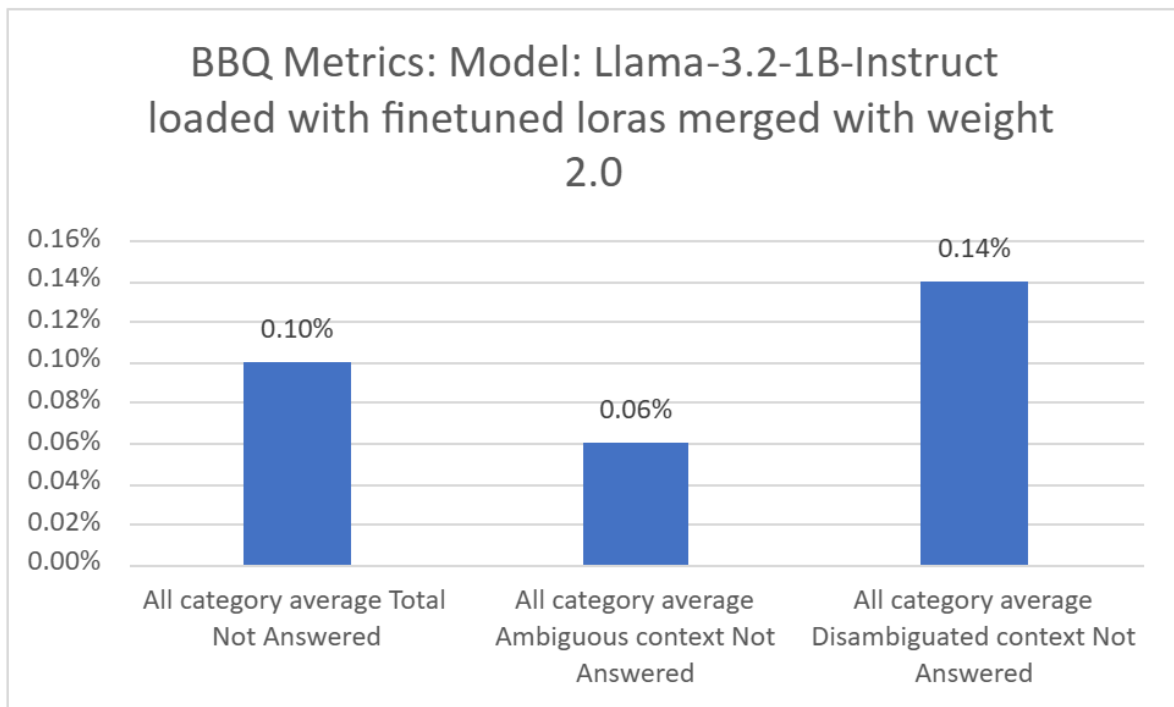
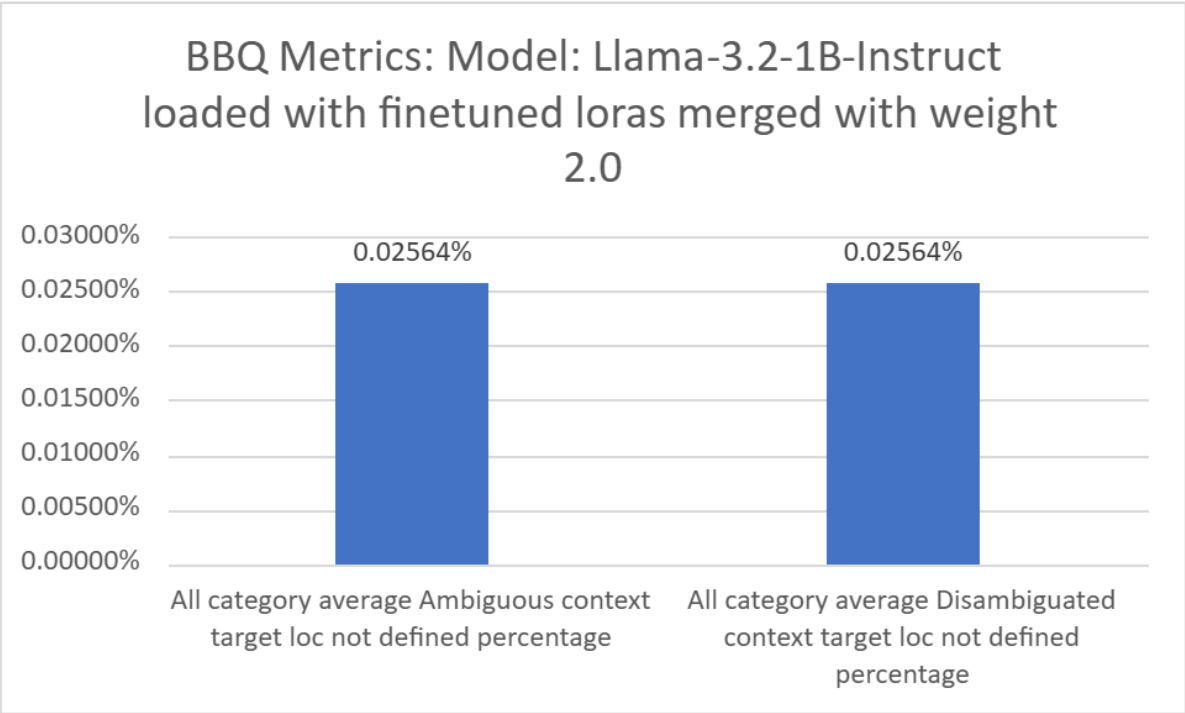


Figure 13: BBQ target loc not defined percentage for model loaded with finetuned LoRas merged with weight 2.0



5.2 MMLU Test Result

To confirm that REFTL preserves general academic domain knowledge without causing catastrophic forgetting, we evaluated all checkpoints on the MMLU (Hendrycks et al., 2020) benchmark.

Table 3: MMLU All-Category Average Metrics.

Model Configuration	Average Accuracy (%)	Not Answered Percentage (%)
Base Model (Llama-3.2-1B-instruct)	36.61	0.88
REFTL (Merge Weight 1.0)	36.80	0.88
REFTL (Merge Weight 2.0)	36.55	1.60

Analysis of MMLU Capabilities

- Accuracy Preservation:** The baseline model achieved an MMLU accuracy of **36.61%**. Merging adapters with weight **1.0** slightly improved overall accuracy to **36.80%**, while weight **2.0** maintained a nearly identical accuracy of **36.55%**.
- Robustness against Catastrophic Forgetting:** These empirical metrics validate that incorporating DARE-TIES adapter merging successfully filters parameter interference. REFTL injects specialized debiasing alignment without degrading underlying language understanding or general academic competencies.

5.2.1 MMLU Test Result for base model

Figure 14: MMLU test accuracy for base model

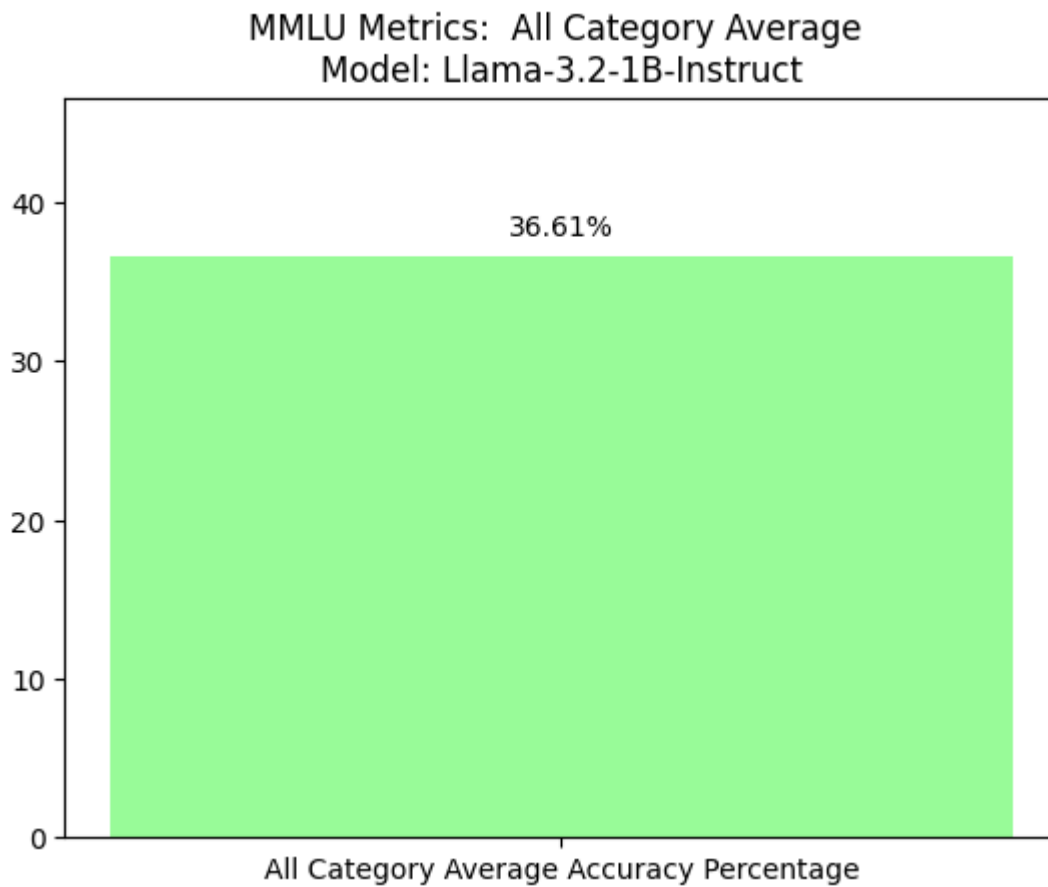
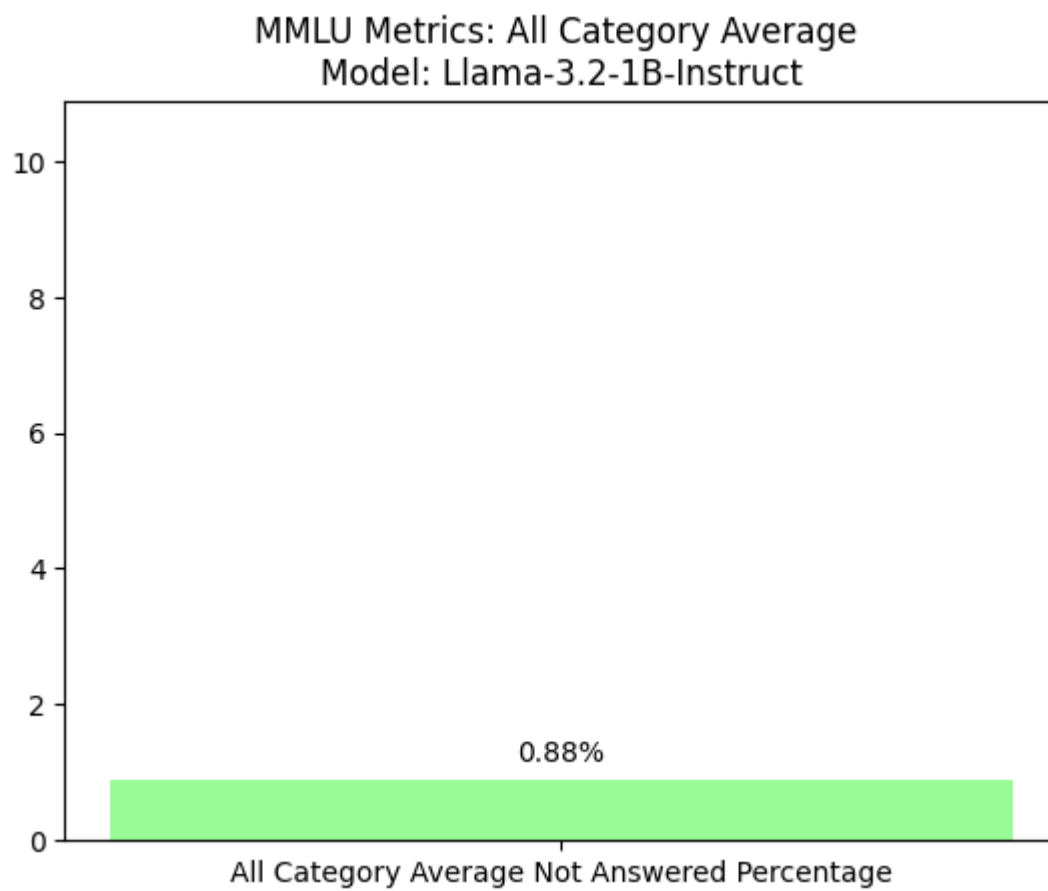


Figure 15: MMLU test not answered percentage for base model



5.2.2 MMLU Test Result for model loaded with merged LoRa adapters with 1.0 weight

Figure 16: MMLU test accuracy for model loaded with finetuned LoRas merged with weight 1.0

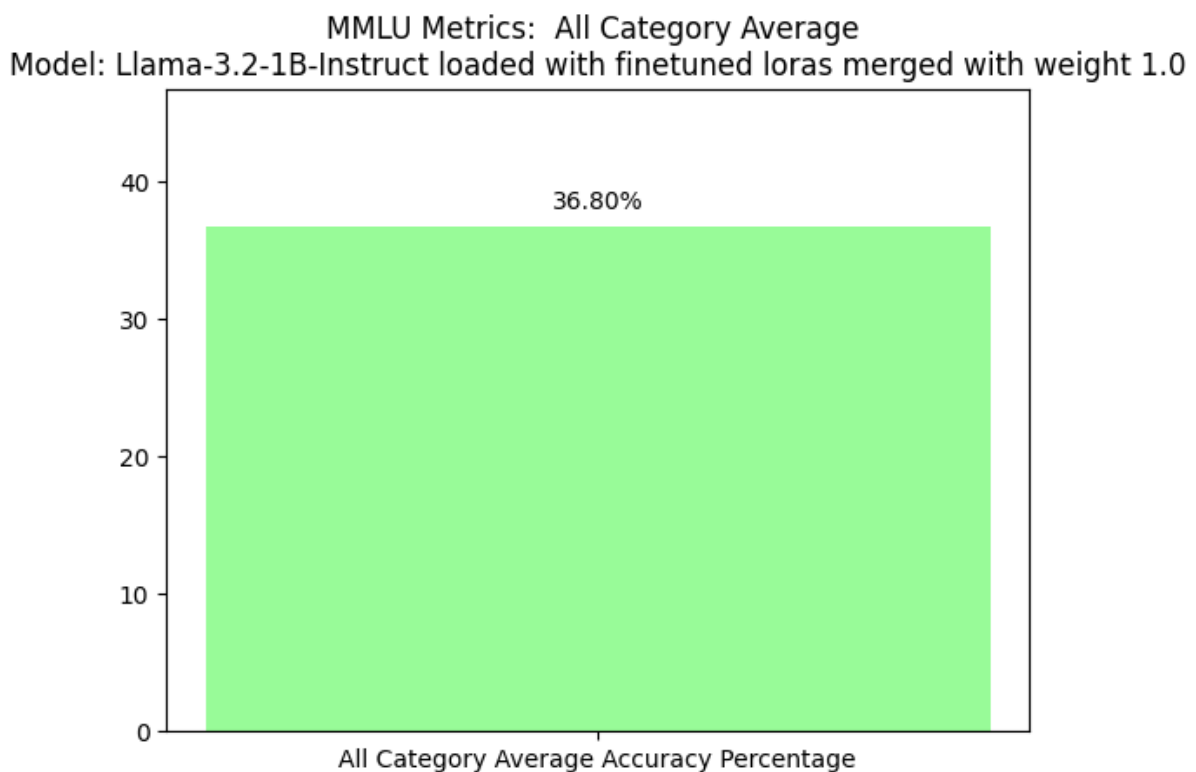
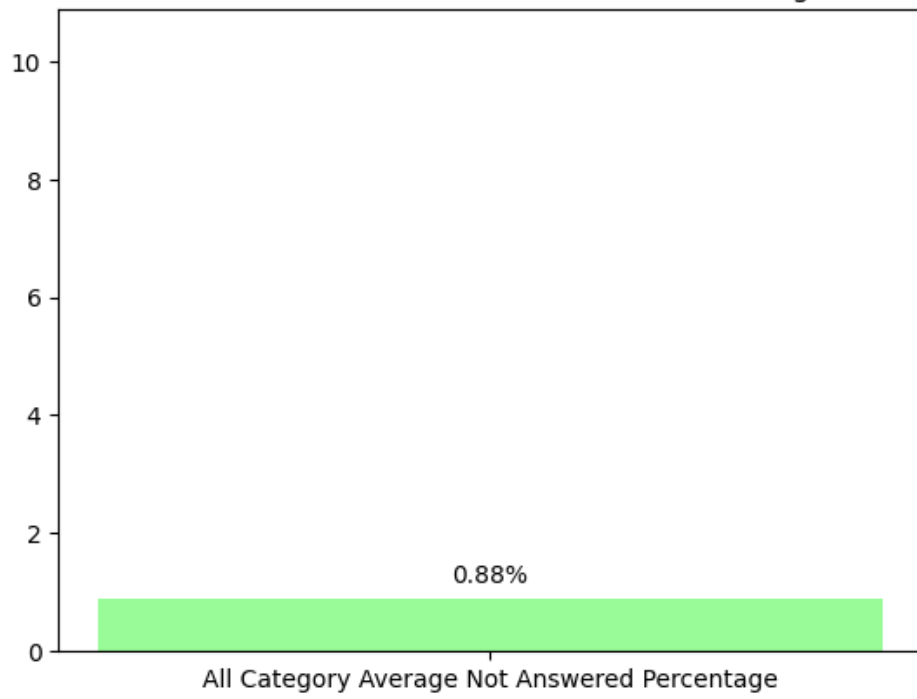


Figure 17: MMLU test unanswered percentage for model loaded with finetuned LoRas merged with weight 1.0

MMLU Metrics: All Category Average
Model: Llama-3.2-1B-Instruct loaded with finetuned loras merged with weight 1.0



5.2.3 MMLU Test Result for model loaded with merged LoRa adapters with 2.0 weight

Figure 18: MMLU test accuracy for model loaded with finetuned LoRas merged with weight 2.0

MMLU Metrics: All Category Average
Model: Llama-3.2-1B-Instruct loaded with finetuned loras merged with weight 2.0

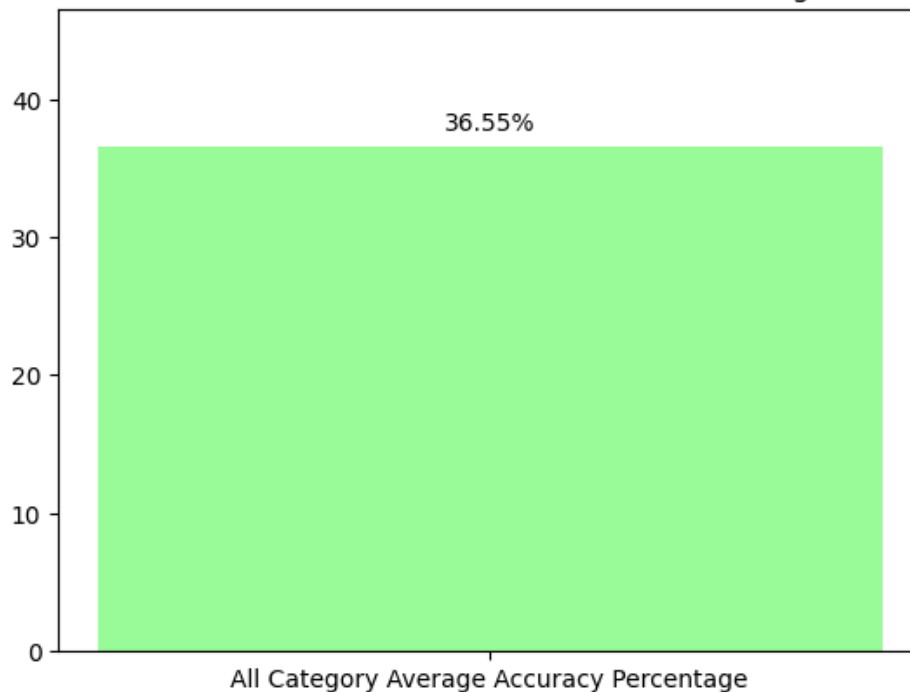
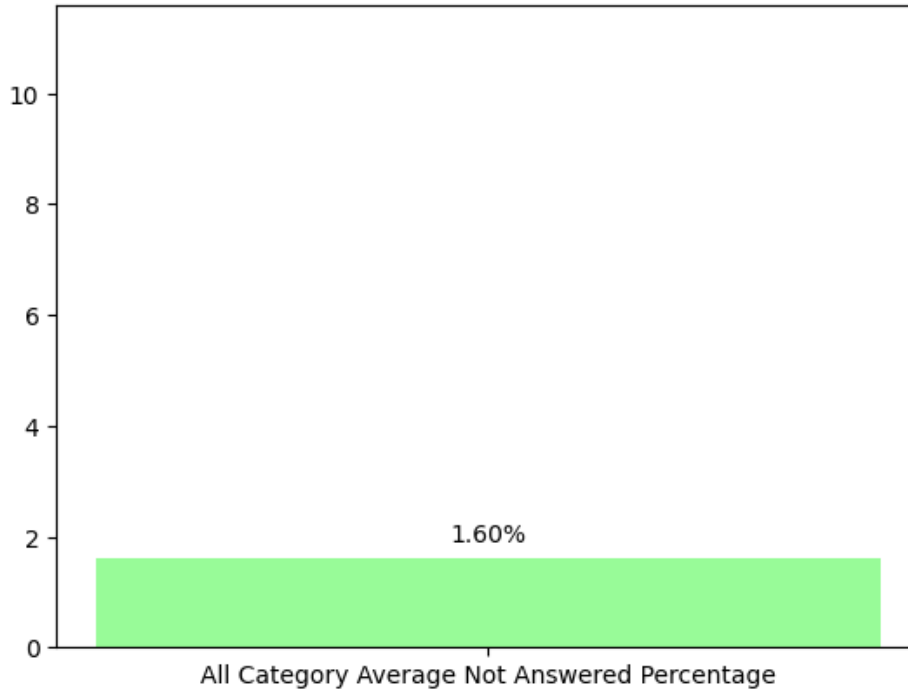


Figure 19: MMLU test unanswered percentage for model loaded with finetuned LoRas merged with weight 2.0

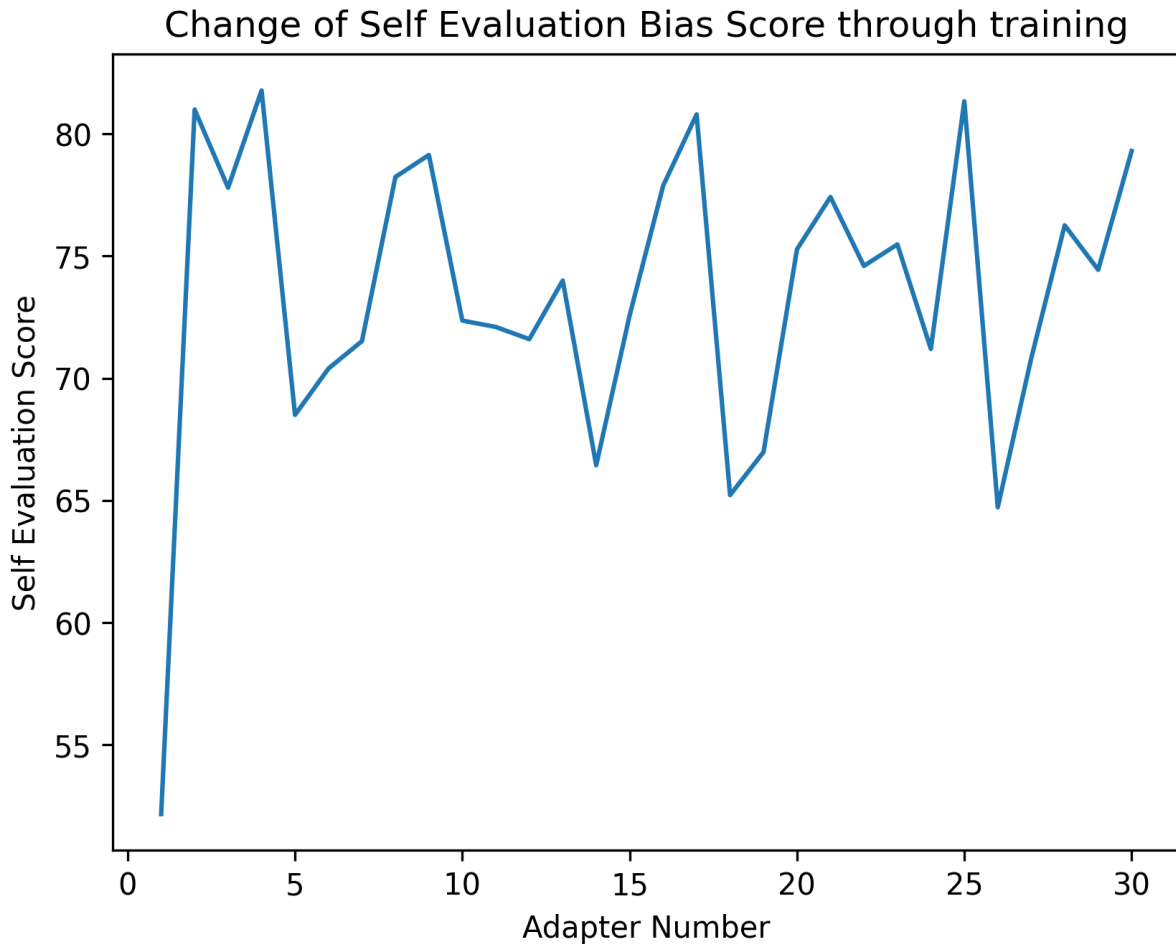
MMLU Metrics: All Category Average
Model: Llama-3.2-1B-Instruct loaded with finetuned loras merged with weight 2.0



5.3 Change of Self evaluation bias score through training

Figure 20 illustrates the dynamic trajectory of the internal Self-Evaluation Bias Score across 30 parallel fine-tuned adapters (generated across 10 outer loops comprising three parallel evolutionary nodes each). The score was quantified using large language model (LLM) self-evaluations bounded within a 0–100 range. Outputs exceeding a numerical density threshold of 0.17 were assigned a score of zero. The number density is the total number of numbers in the text divided by the length of text. The plotted curve depicts the mean score calculated at each evaluation round.

Figure 20: Change of Self evaluation bias score through training



Trajectory Dynamics:

- 1. Initial Acceleration Phase:** The initial base adapter (Adapter 1) records a low self-evaluation score of under 55.0. Upon introducing debiasing persona guidance and self-critique penalties, the score jumps rapidly above 80.0 within the first few evolutionary updates
- 2. Evolutionary Exploration & Selection:** In subsequent iterations (Adapters 5 through 30), the score fluctuates dynamically within a high-performing band of ~65 to ~81. These oscillations reflect the active evolutionary branching mechanism: local parallel nodes generate diverse dataset variations, and lower-scoring branch trajectories are subsequently pruned or re-merged via DARE-TIES.

3. **Convergence to Robust Baseline:** Across extended training, the self-evaluation score maintains an elevated baseline compared to initial initialization. This ongoing iterative refinement ensures that the cumulative champion adapter pool continually selects high-quality alignment updates while avoiding local minima convergence.

6. Discussions and Limitations

6.1 Why DARE-TIES Model merging prevents degradation

Continuous fine-tuning usually suffers from catastrophic forgetting (Zhai et al., 2023) where updating weights on task A degrades performance on task B. **REFTL** avoids this by freezing the base model parameter and performing evolutionary operations only on LoRA (Hu et al., 2021) parameters.

By applying DARE (Yu et al., 2023) operation, a high percentage of low-rank updates are dropped, leaving only the most critical, consensus based parameter changes. The TIES (Yadav et al., 2023) operation then resolves sign conflict before updating the weights. This keeps the adapter representation clean and sparse, allowing the model to acquire debiasing alignment while preserving the capabilities of the base model.

6.2 Limitations

- **Self-Evaluation Bias:** **REFTL** relies on the model itself to evaluate and grade generated responses. While our framework filters out bad generation by filtering out low self evaluation score content and content with high number density, there remains a risk of “echo-chamber” optimization, where the model prioritizes responses that match its internal prior evaluation metrics rather than general objective fairness.
- **High Computation requirement:** Running 3 evolutionary branches in parallel over 10 outer loops requires training 30 separate QLoRA (Dettmers et al., 2023) adapters. Although

QLoRA (Dettmers et al., 2023) is efficient, this framework requires substantial GPU time.

7. Conclusion

We introduced Reinforcement learning via Evolutionary Fine-Tuning Loops (**REFTL**), an automated, closed-loop framework for aligning Large Language Models without human annotation. By integrating evolutionary selection over parallelized QLoRA (Dettmers et al., 2023), and QLoRA merging via DARE-Ties (Yadav et al., 2023; Yu et al., 2023), **REFTL** achieves bias reduction.

Evaluations on the BBQ (Parrish et al., 2022) benchmark demonstrate that the **REFTL** framework successfully reduces overall bias while there is little drop in disambiguated context accuracy as alignment tax. **REFTL** simultaneously maintains general reasoning capabilities on the MMLU (Hendrycks et al., 2020) benchmark. Future research will focus on extending the **REFTL** framework on other LLM models.

Reference

Dan Hendrycks, Collin Burns, Steven Basart, Andy Zou, Mantas Mazeika, Dawn Song, Jacob Steinhardt (2020). Measuring Massive Multitask Language Understanding. **arXiv** <https://arxiv.org/abs/2009.03300>

Alicia Parrish, Angelica Chen, Nikita Nangia, Vishakh Padmakumar, Jason Phang, Jana Thompson, Phu Mon Htut, Samuel R. Bowman (2022). BBQ: A hand-built bias benchmark for question answering. *Findings of the Association for Computational Linguistics: ACL 2022*, 2086-2105. <https://aclanthology.org/2022.findings-acl.165/>

Rafael Rafailov, Archit Sharma, Eric Mitchell, Stefano Ermon, Christopher D. Manning, Chelsea Finn (2023). Direct Preference Optimization: Your Language Model is Secretly a Reward Model. **arXiv** <https://arxiv.org/abs/2305.18290>

Prateek Yadav, Derek Tam, Leshem Choshen, Colin Raffel, Mohit Bansal (2023). TIES-Merging: Resolving Interference When Merging Models. **arXiv** <https://arxiv.org/abs/2306.01708>

Le Yu, Bowen Yu, Haiyang Yu, Fei Huang, Yongbin Li (2023). Language Models are Super Mario: Absorbing Abilities from Homologous Models as a Free Lunch. **arXiv** <https://arxiv.org/abs/2311.03099>

Long Ouyang, Jeff Wu, Xu Jiang, Diogo Almeida, Carroll L. Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, John Schulman, Jacob Hilton, Fraser Kelton, Luke Miller, Maddie Simens, Amanda Askell, Peter

Welinder, Paul Christiano, Jan Leike, Ryan Lowe (2022) Training language models to follow instructions with human feedback. **arXiv** <https://arxiv.org/abs/2203.02155>

Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, Weizhu Chen (2021) LoRA: Low-Rank Adaptation of Large Language Models **arXiv** <https://arxiv.org/abs/2106.09685>

Tim Dettmers, Artidoro Pagnoni, Ari Holtzman, Luke Zettlemoyer (2023) QLoRA: Efficient Finetuning of Quantized LLMs. **arXiv** <https://arxiv.org/abs/2305.14314>

Yizhong Wang, Yeganeh Kordi, Swaroop Mishra, Alisa Liu, Noah A. Smith, Daniel Khashabi, Hannaneh Hajishirzi (2022) Self-Instruct: Aligning Language Models with Self-Generated Instructions. **arXiv** <https://arxiv.org/abs/2212.10560>

Yuntao Bai, Saurav Kadavath, Sandipan Kundu, Amanda Askell, Jackson Kernion, Andy Jones, Anna Chen, Anna Goldie, Azalia Mirhoseini, Cameron McKinnon, Carol Chen, Catherine Olsson, Christopher Olah, Danny Hernandez, Dawn Drain, Deep Ganguli, Dustin Li, Eli Tran-Johnson, Ethan Perez, Jamie Kerr, Jared Mueller, Jeffrey Ladish, Joshua Landau, Kamal Ndousse, Kamile Lukosuite, Liane Lovitt, Michael Sellitto, Nelson Elhage, Nicholas Schiefer, Noemi Mercado, Nova DasSarma, Robert Lasenby, Robin Larson, Sam Ringer, Scott Johnston, Shauna

Kravec, Sheer El Showk, Stanislav Fort, Tamera Lanham, Timothy Telleen-Lawton, Tom Conerly, Tom Henighan, Tristan Hume, Samuel R. Bowman, Zac Hatfield-Dodds, Ben Mann, Dario Amodei, Nicholas Joseph, Sam McCandlish, Tom Brown, Jared Kaplan (2022) Constitutional AI: Harmlessness from AI Feedback. **arXiv** <https://arxiv.org/abs/2212.08073>

Yuexiang Zhai, Shengbang Tong, Xiao Li, Mu Cai, Qing Qu, Yong Jae Lee, Yi Ma (2023) Investigating the Catastrophic Forgetting in Multimodal Large Language Models. **arXiv** <https://arxiv.org/abs/2309.10313>

A. Z. Broder (1997) On the resemblance and containment of documents. <https://ieeexplore.ieee.org/document/666900>

Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, et al (2024) The Llama 3 Herd of Models. **arXiv** <https://arxiv.org/abs/2407.21783>

Gabriel Ilharco, Marco Tulio Ribeiro, Mitchell Wortsman, Suchin Gururangan, Ludwig Schmidt, Hannaneh Hajishirzi, Ali Farhadi (2022) Editing Models with Task Arithmetic **arXiv** <https://arxiv.org/abs/2212.04089>

Zixiang Chen, Yihe Deng, Huizhuo Yuan, Kaixuan Ji, Quanquan Gu (2024) Self-Play Fine-Tuning Converts Weak Language

Models to Strong Language Models **arXiv**

<https://arxiv.org/abs/2401.01335>

Katherine Lee, Daphne Ippolito, Andrew Nystrom, Chiyuan Zhang, Douglas Eck, Chris Callison-Burch, Nicholas Carlini (2021) Deduplicating Training Data Makes Language Models Better **arXiv** <https://arxiv.org/abs/2107.06499>

Amanda Askell, Yuntao Bai, Anna Chen, Dawn Drain, Deep Ganguli, Tom Henighan, Andy Jones, Nicholas Joseph, Ben Mann, Nova DasSarma, Nelson Elhage, Zac Hatfield-Dodds, Danny Hernandez, Jackson Kernion, Kamal Ndousse, Catherine Olsson, Dario Amodei, Tom Brown, Jack Clark, Sam McCandlish, Chris Olah, Jared Kaplan (2021) A General Language Assistant as a Laboratory for Alignment **arXiv** <https://arxiv.org/abs/2112.00861>

Ganqu Cui, Lifan Yuan, Ning Ding, Guanming Yao, Bingxiang He, Wei Zhu, Yuan Ni, Guotong Xie, Ruobing Xie, Yankai Lin, Zhiyuan Liu, Maosong Sun (2023) UltraFeedback: Boosting Language Models with Scaled AI Feedback **arXiv** <https://arxiv.org/abs/2310.01377>

Takuya Akiba, Makoto Shing, Yujin Tang, Qi Sun, David Ha (2024) Evolutionary Optimization of Model Merging Recipes **arXiv** <https://arxiv.org/abs/2403.13187>

Mitchell Wortsman, Gabriel Ilharco, Samir Yitzhak Gadre, Rebecca Roelofs, Raphael Gontijo-Lopes, Ari S. Morcos, Hongseok Namkoong, Ali Farhadi, Yair Carmon, Simon Kornblith, Ludwig Schmidt (2022) Model soups: averaging

weights of multiple fine-tuned models improves accuracy
without increasing inference time **arXiv**

<https://arxiv.org/abs/2203.05482>

Weizhe Yuan, Richard Yuanzhe Pang, Kyunghyun Cho, Xian Li,
Sainbayar Sukhbaatar, Jing Xu, Jason Weston (2024) Self-
Rewarding Language Models **arXiv**

<https://arxiv.org/abs/2401.10020>

Harrison Lee, Samrat Phatale, Hassan Mansoor, Thomas
Mesnard, Johan Ferret, Kellie Lu, Colton Bishop, Ethan Hall,
Victor Carbune, Abhinav Rastogi, Sushant Prakash (2023)
RLAIF vs. RLHF: Scaling Reinforcement Learning from Human
Feedback with AI Feedback **arXiv**

<https://arxiv.org/abs/2309.00267>